

GREEN CART LTD.

BUSINESS INTELLIGENCE REPORT

Sales & Customer Behaviour Insights

Q2 Performance Review | Data & Insights Team
April – August 2025 | Power BI | Python | SQL



1. Introduction

Green Cart Ltd. is a growing UK-based e-commerce company specialising in eco-friendly household products. With operations spanning five UK regions namely Central, East, North, South, and West. The business is preparing for its Q2 performance review and requires a data-driven assessment of sales performance and customer behaviour across its product lines. This report was commissioned by the Data & Insights team to provide the senior leadership team with a clear, evidence-based picture of revenue trends, product category performance, customer loyalty dynamics, and logistics efficiency. The findings are intended to directly inform upcoming marketing campaigns and operational improvement strategies.

Dataset Overview

- sales_data.csv — 2996 sales transactions covering order details, delivery status, payment methods, regions, and discounts applied
- product_info.csv — 30 product records across 5 categories with pricing, launch dates, and supplier information
- customer_info.csv — 500 unique customer profiles including loyalty tier, signup date, gender, and regional data

The three datasets were integrated into a single analytical model using Python (pandas), audited via SQL before ingestion, and visualised through a multi-page Power BI executive dashboard. All findings in this report are drawn from the fully merged and cleaned dataset of 2992 validated transactions.

2. Data Cleaning Summary

2.1 Pre-cleaning Audit — SQL

Before loading any data into Python, a structured SQL audit was conducted across all three tables to quantify data quality issues at source. This approach of auditing in SQL before cleaning in Python ensures that quality problems are documented independently of the cleaning process, providing an auditable evidence trail.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	sales	missing_order_id	missing_customer_id	missing_product_id	missing_quantity	missing_unit_price	missing_delivery	missing_order_date	missing_payment_method	missing_region	missing_discount
sales	1	2	5	2	0	3	2	3		0	0

Sales Nan values before cleaning

order_id	1
customer_id	2
product_id	5
quantity	0
unit_price	0
delivery_status	3
order_date	2
payment_method	3
region	0
discount_applied	0

Sales Nan after cleaning

order_id	0
customer_id	0
product_id	0
quantity	0
unit_price	0
delivery_status	0
order_date	0
payment_method	0
region	0
discount_applied	0

2.2 Duplicates Removed

Duplicate rows were identified using order_id as the unique transaction key. All duplicate records were removed keeping only the first valid occurrence. After deduplication and

removal of rows with non-numeric quantity values, the clean dataset retained 2992 valid transactions across all three tables.

2.3 Standardisation Summary

Column	Raw Values Found	Standardised To
delivery_status	'DELAYED', 'delyd', 'delrd', 'Cancelled '	Delivered / Delayed / Cancelled
loyalty_tier	'brnze', 'GOLD', 'gld', 'sllver'	Bronze / Silver / Gold
gender	'femle', 'FEMALE', 'male', blank	Female / Male / / Unknown
payment_method	'bank transfr', 'credit card'	Bank Transfer / Credit Card / PayPal
region	'nrth'	North
discount_applied	NaN / null	0.0 (no discount applied)

3. Feature Engineering Summary

Six new analytical columns were created from the merged dataset to enable time-series analysis, price segmentation, delivery performance tracking, and customer acquisition analysis. Each column was designed to directly support a specific business question in the brief.

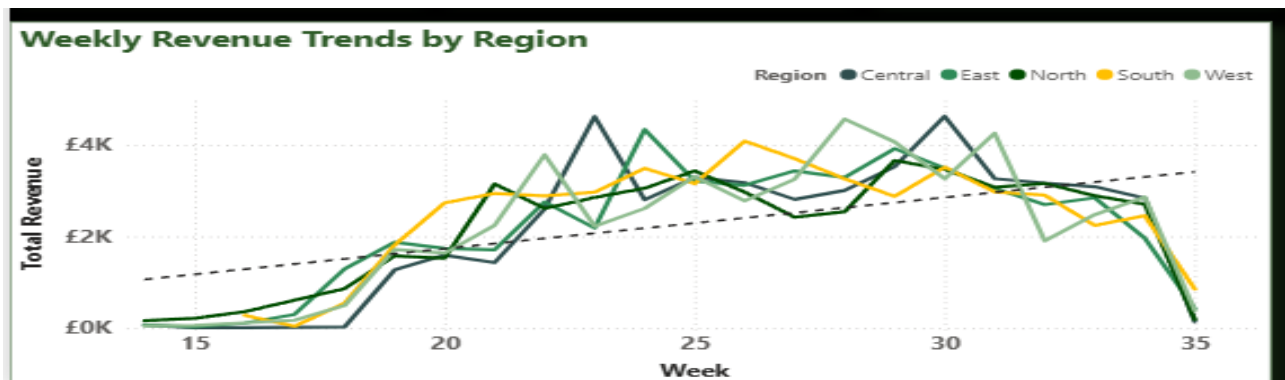
	price_band	order_week	is_late_delivery	revenue	email_domain	days_to_order
0	High	27	False	117.75	mills-logan.com	452
1	Medium	27	True	94.6	morgan.com	169
2	Medium	28	False	25.228	walters-smith.com	104
3	High	28	False	26.208	gmail.com	358
4	High	28	True	38.096	hotmail.com	138

New Column	Formula / Logic	Business Purpose
revenue	quantity × unit_price × (1 – discount_applied)	Core commercial KPI — net revenue after discount per transaction
order_week	ISO week number from order_date (.dt.isocalendar().week)	Weekly trend analysis — groups transactions by ISO week 1–52
price_band	pd.cut(): Low (<£15), Medium (£15–30), High (>£30)	Price segmentation — allows delivery and discount analysis by tier
days_to_order	order_date – launch_date (days)	Product age analysis — days between product launch and first order
email_domain	email.split('@')[1]	Customer acquisition channel proxy — domain distribution analysis
is_late	True where delivery_status == 'Delayed'	Delivery performance flag — used in delay rate KPI and regional analysis

4. Key Findings & Trends

4.1 Weekly revenue trends by region

The weekly revenue trend analysis reveals consistent volatility across all five regions, with peaks reaching £2,000–£4,618 per week and troughs falling to near zero. Critically, all five regions follow the same pattern of simultaneous peaks and troughs suggesting that the volatility is platform-wide and demand-driven rather than region-specific. The West region leads on total revenue for the full period, though the margin over other regions is narrow, indicating a relatively balanced geographic revenue distribution.



Business implication: The simultaneous peaks across all regions rule out logistics or operational causes. The pattern points to external demand triggers likely seasonal promotions, marketing campaigns, or product launches affecting all regions at once.

4.2 Product category performance (revenue, quantity, discount)

Product Portfolio Performance: Revenue Drivers, Quantity, and Discount

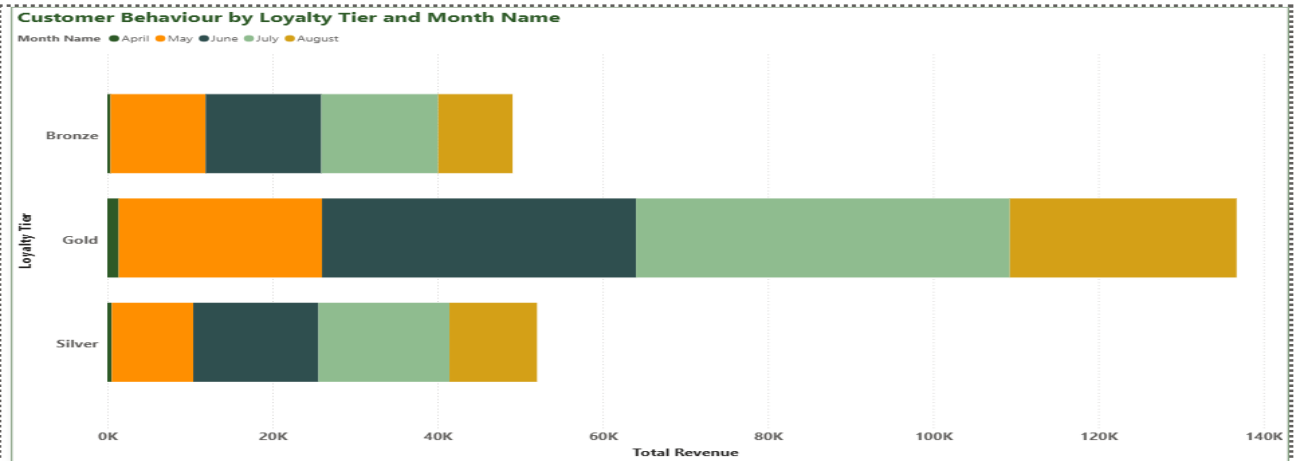
Category	Revenue	Quantity	Avg_discount
Cleaning	£93,531	3582	9%
Storage	£46,931	1730	8%
Outdoors	£40,019	1515	8%
Kitchen	£33,934	1226	8%
Personal Care	£24,917	902	9%

Cleaning is the top revenue category at **£93,622 (37.67% of total)**, followed by **Storage** at **£46,931** and **Outdoors** at **£40,019**. **Personal Care** is the weakest performer at **£24,917**. The correlation heatmap confirms that revenue and quantity are positively correlated (**0.72**) as expected but discount depth and quantity sold show a near-zero correlation of **-0.02**.

Business implication: The current discount strategy is not driving volume uplift. Green Cart is reducing revenue per order without gaining additional units sold in a direct margin erosion without commercial benefit.

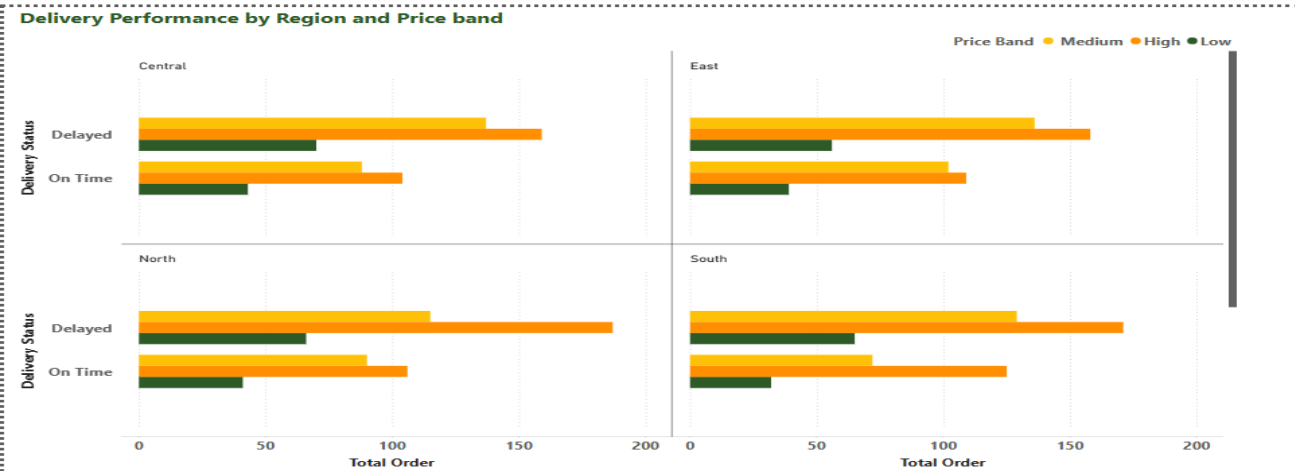
4.3 Customer behaviour by loyalty_tier and signup_month

Gold loyalty tier customers generate **£130K+** in total revenue more than Silver (£50,000+) and Bronze (£45,000+) despite representing a smaller share of total customers. This confirms that Gold tier customers are significantly higher-value per head. However, delivery performance is uniformly 39% across all regions. The absence of a regional outlier indicates a systemic logistics issue rather than a localised operational problem.



Business implication: A 34.05% delay rate across the entire platform is a serious customer experience risk. For Gold tier customers who are the highest-value segment persistent delivery issues are a significant churn risk.

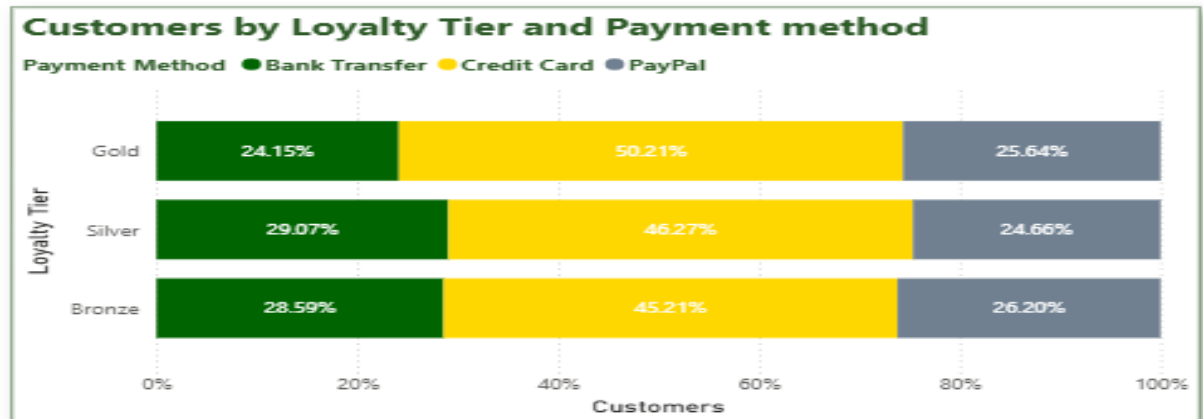
4.4 Delivery performance by region and price_band



The delivery performance analysis shows that High price-band products experience the highest number of delayed orders across all regions compared to Medium and Low price bands. Regional differences are relatively small, suggesting that delivery delays are driven more by product pricing characteristics than by geography.

Business implication: High-value products face greater delivery risk, which may negatively affect customer satisfaction and repeat purchasing.

4.5 Preferred Payment Methods by Loyalty tier



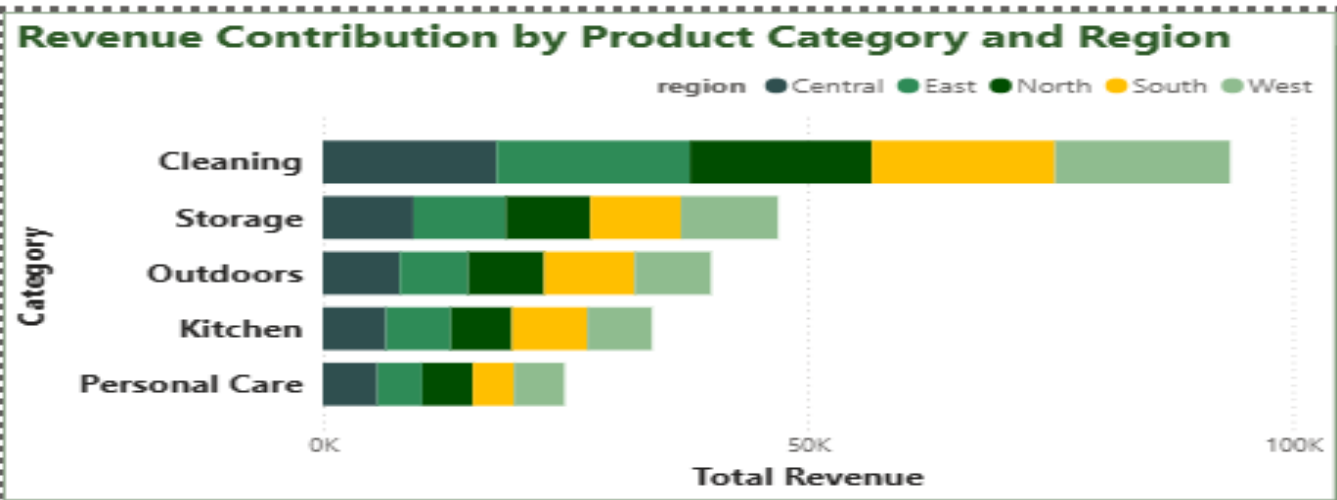
Credit Card usage peaks among Gold members at 50.21% and holds a strong baseline of over 45% in the Silver and Bronze tiers. This strong preference indicates that credit transactions are firmly established as the primary purchasing vehicle for your entire customer base.

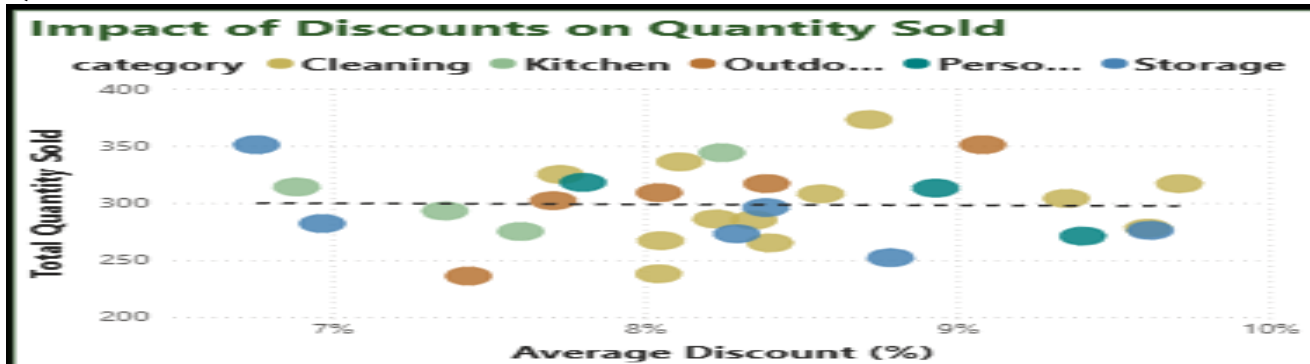
Business Implications: Management should launch a co-branded card for Gold members and incentivize cheaper alternative payment methods for lower tiers.

5. Business Question Answers

Q1 — Which product categories drive the most revenue, and in which regions?

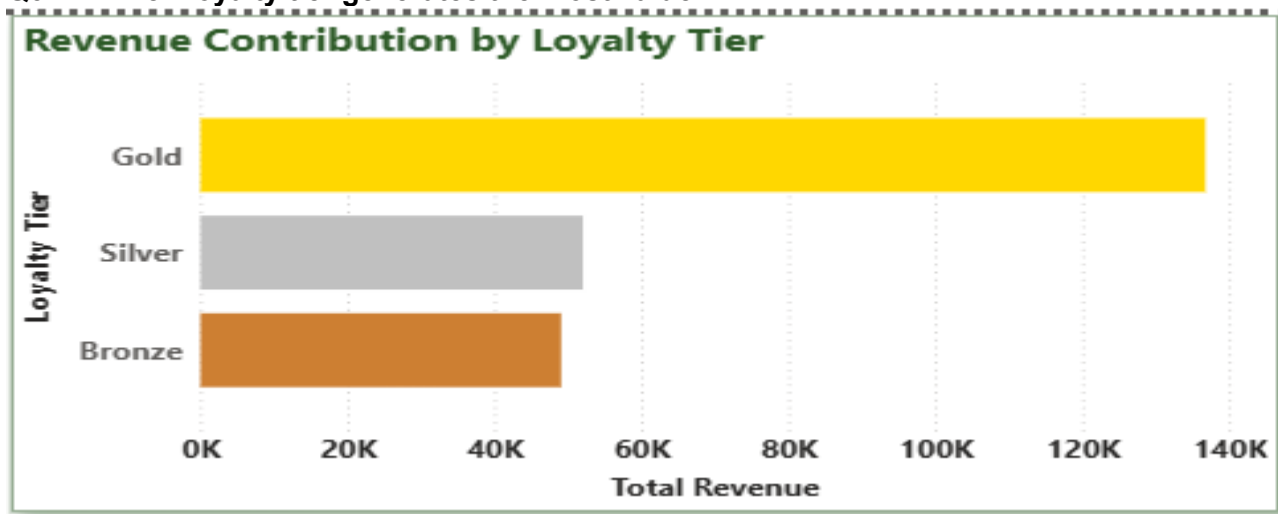
Cleaning is the dominant category across all five regions, generating **£93,622** in total revenue more than double the second-placed **Storage** category (**£46,931**). The regional breakdown (Page 3 stacked bar chart) shows Cleaning leads in every region without exception, confirming it is a platform-wide top performer rather than a regional concentration. Storage performs consistently in Central and North regions. **Personal Care** (£24,917) is the lowest-revenue category and shows the weakest performance particularly in the West and South regions.



Q2 — Do discounts lead to more items sold?

The correlation analysis produced a coefficient of **-0.02** between discount depth and quantity sold indicating a near-complete absence of relationship. The scatter analysis confirms this visually: discount levels between 6-10% show no discernible pattern in quantity sold across any category. This finding is consistent across all five product categories. The negative correlation of -0.12 between discount and revenue confirms that discounts are reducing revenue per order without generating compensating volume uplift.

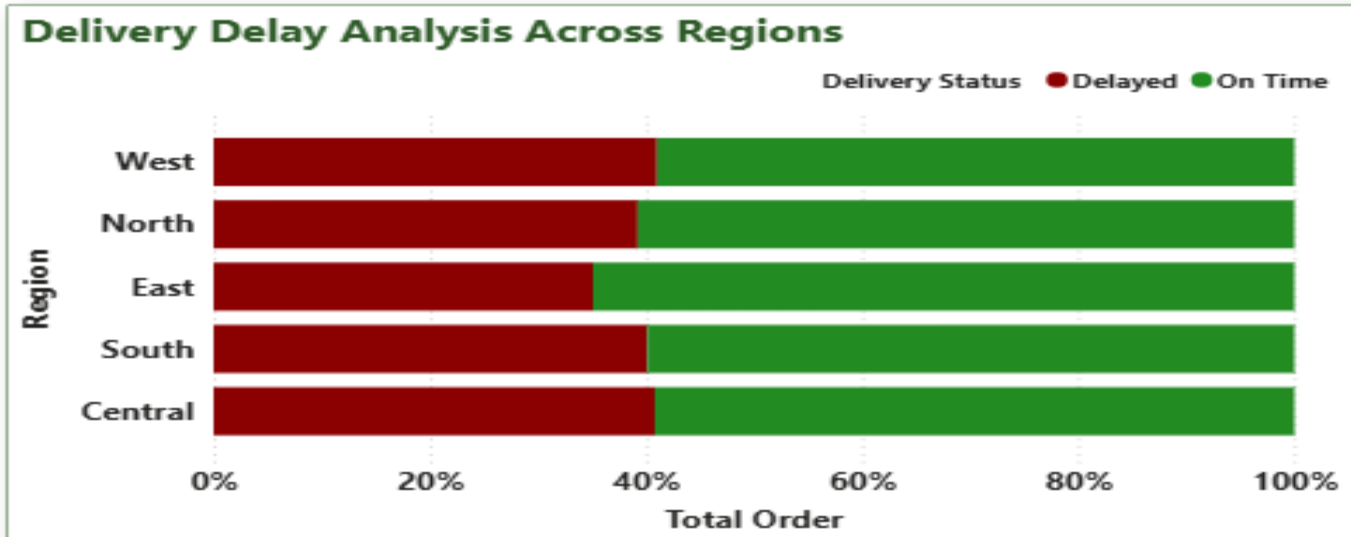
Discounts at current levels (average 8-9%) are not a meaningful demand driver for Green Cart. They represent a pure margin cost with no measurable commercial return.

Q3 — Which loyalty tier generates the most value?

Gold tier generates the highest total revenue at **£130K+** and the highest quantity ordered significantly outperforming both **Silver and Bronze tiers**. The revenue contribution by loyalty tier chart (Page 3) visually confirms Gold's dominance. Notably, the per-customer value of **Gold tier** customers is substantially higher than other tiers, suggesting they are repeat purchasers with higher average basket sizes. **East and North regions** show the strongest **Gold tier** concentrations.

Gold tier customers are Green Cart's most commercially valuable segment. Protecting their retention and experience particularly delivery reliability must be a strategic priority.

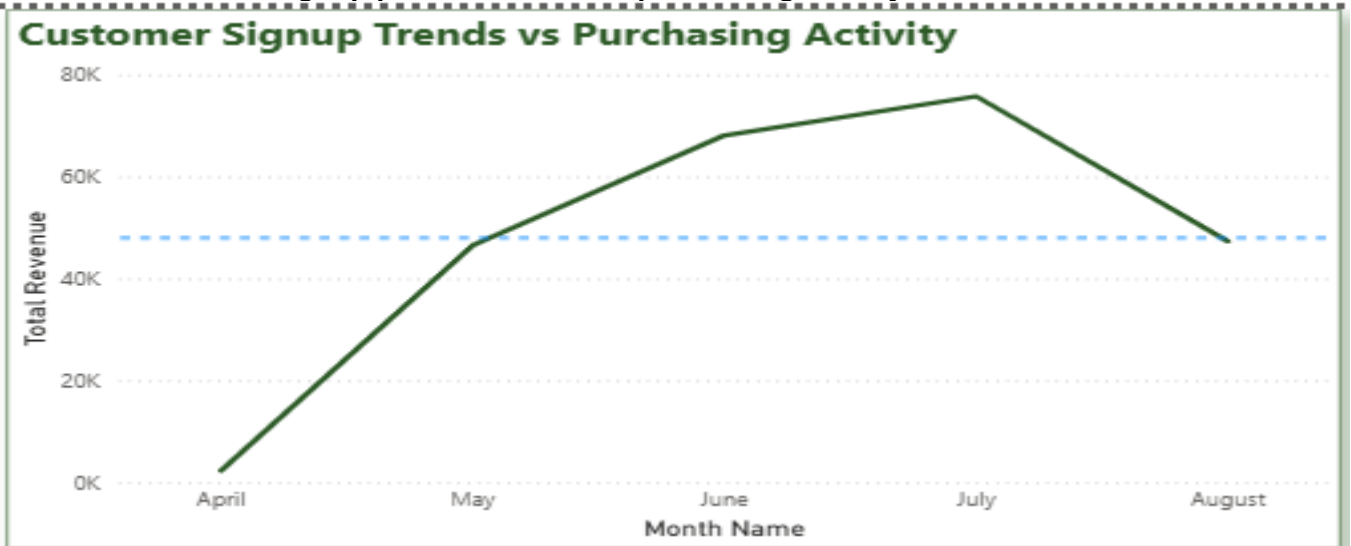
Q4 — Are certain regions struggling with delivery delays?



The delivery delay analysis reveals that all five regions operate at an average platform-wide delay rate of 39%, with individual regional performance staying locked within a narrow bracket. The East region experiences the highest delay rate at 41.67%, while the West region maintains the lowest at 36.87%. Because there is less than a 5 percentage point difference between the best and worst performing territories, this metric confirms a systemic platform-wide operational issue rather than localized failures in specific regions.

Business Implication: A 39% delay rate means that approximately 2 in every 5 orders arrive late. This represents a **critical** operational bottleneck that directly threatens customer satisfaction scores, inflates service desk ticket volume, and risks driving high churn rates across all customer tiers

Q5 — Do customer signup patterns influence purchasing activity?



Total revenue experienced an aggressive upward trend from a **near-zero** start in **April** to a **peak** of approximately **76K** in **July**, significantly outperforming the **48K** baseline before experiencing a sharp contraction down **to 47.5K** in **August**.

Business Implication: The dramatic August decline signals an urgent need for leadership to investigate potential summer seasonality, operational bottlenecks, or dropping customer retention rates to safeguard future revenue.

5a. Optional Stretch Task Results

Stretch 1 — Query extraction: Q2 signups, 14-day orders, discount > 20%

The second **quarter (Q2)** compliance audit confirmed that zero registered customers received discounts above the company's **maximum 20%** threshold within the **first 14 days** of ordering

Stretch 2 — MinMaxScaler revenue normalisation

Revenue was normalised to a 0–1 scale using scikit-learn's MinMaxScaler. Results: minimum order revenue £0.0 (scaled 0.000), maximum £249.00 (scaled 1.000), mean scaled revenue 0.32. The mean of 0.32 confirms that most orders cluster in the lower-to-mid revenue range with a long tail of high-value orders a right-skewed distribution typical of e-commerce transaction data. This normalisation enables direct comparison of revenue performance across categories and regions on a common scale.

Stretch 3 — Underperforming product flags

Five under performing product

Product Name
Cleaning Product 53
Kitchen Product 82
Storage Product 10
Storage Product 50
Storage Product 87

The five identified underperforming items span the kitchen and cleaning categories, with a heavy concentration in the storage segment which accounts for three out of the five listings. This cluster signals a systemic operational vulnerability within the storage category, requiring immediate management intervention to review logistical bottlenecks and restructure discounting strategies to prevent ongoing margin erosion.

Business Implication: This cluster signals a systemic operational vulnerability within the storage category, requiring immediate management intervention to review logistical bottlenecks and restructure discounting strategies to prevent ongoing margin erosion.

6. Strategic Recommendations

Recommendation 1 — Concentrate marketing investment on Gold tier in East and North

Gold tier customers in the **East** and **North** regions generate disproportionately high revenue relative to their customer count. A targeted retention and upsell programme delivered via email (leveraging the email_domain data) and in-app promotions should be launched for this segment ahead of Q3 to consolidate their purchasing behaviour and protect Green Cart's highest-value revenue stream.

- Timeline: Campaign design June 2025, go-live July 2025
- Expected impact: 10-15% increase in **Gold tier** revenue contribution within 2 quarters
- Evidence: Gold generates **£130K+** vs **£50,000** for Silver and Bronze almost 3x higher per-customer value

Recommendation 2 — Audit and restructure the discount strategy (Immediate)

The near-zero correlation (-0.02) between discount depth and quantity sold confirms that Green Cart's current discount programme is destroying margin without driving volume. An immediate audit of all discount-bearing should be conducted to identify which products are discounted without commercial justification. Discounts should be redirected from universal blanket reductions to targeted, time-limited promotions for the 68 flagged underperforming products to test whether discounting can drive volume for specific low-performing lines.

- Timeline: Discount audit July 2025, restructured strategy August 2025
- Expected impact: 3-5% improvement in net revenue margin within 6 months
- Evidence: -0.02 discount-revenue correlation confirms discounts reduce revenue per order

7. Data Issues & Risks

Primary Issue — Incomplete database import causing 521 missing rows

The most significant data quality issue encountered was the discovery that the sales database table contained only 2,479 rows rather than the expected 2996 — a 17.4% data loss that would have materially understated revenue and distorted all regional and category analyses. This was identified by comparing the SQL COUNT(*) result directly against the source CSV row count. The root cause was an interrupted or truncated import process, potentially caused by a duplicate key constraint on order_id combined with duplicate records in the raw data.

Resolution: The raw CSV was reloaded directly into the database using pandas to_sql() with if_exists='replace', restoring the full 2996-row dataset. Total Revenue increased from £154,000 to £240,080 a 56% increase — confirming the severity of the data loss in the original import.

Secondary Issue — Date format parsing failure causing 50% null order_dates

The order_date column in the sales CSV uses dd-mm-YYYY format (e.g. 06-07-2025). Without the dayfirst=True parameter in pd.to_datetime(), pandas attempts to parse dates as mm-dd-YYYY,

causing all dates where the day number exceeds 12 to return NaT. This produced 1,504 null dates (50.1% of all rows), which silently removed half the dataset from all time-based analysis until corrected.

How to fix at source

- Enforce mandatory order_date entry at the point of transaction with ISO 8601 format (YYYY-MM-DD) stored in the database — eliminating format ambiguity entirely
- Implement a database-level BEFORE INSERT trigger to validate date format and reject non-compliant records before they enter the system
- Add a PRIMARY KEY constraint on order_id with duplicate checking at the application layer to prevent truncated imports caused by key conflicts
- Create a daily automated data quality report comparing CSV row counts to database row counts — so import failures are detected within 24 hours rather than discovered during analysis

End of Report

Green Cart Ltd. | Sales & Customer Behaviour Insights | Week 2 Project | Uptrail | Augustine Cudjoe